

Investigation on the vortex-and-wake-induced vibration of a separated-box bridge girder



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ABSTRACT

Experiments are performed on three spring-supported separated-box girder section models in a closed wind tunnel to investigate the vortex-induced vibration (VIV) characteristics. A particle image velocimetry (PIV) instrument is employed to obtain the flow field around the models. Four VIV-sensitive velocity intervals are observed for the test model, in which vertical and torsional oscillations are repeatedly induced. Comparisons with the control groups show that vortex shedding in the gap is the crucial factor that induces VIVs. Wake-induced vibration (WIV) is observed to occur with VIV in the gap, accompanied by a dramatic magnification in vibrational amplitude. The WIV becomes weaker and disappears as the wind velocity increases, and the wake vortex-shedding mode changes from the classic 2S mode to a 2(2S) mode. The vortex-shedding frequency shows that it is a special lock-in mode of 1:2 phase-locking, which is found by taking the vibrating model and vortex-shedding as two coupled oscillators.

1. Introduction

As bridges main spans increase, the separated-box girder (SBG) is considered for super-long bridges due to its superior flutter stability at high wind speed. However, health monitoring of a suspension SBG bridge (main span: 1650 m) revealed that the girder is prone to unexpected vortex-induced vibration (VIV) at low wind velocity, which is harmful to the bridge life-span and driving comfort (Li et al., 2011, 2014b). The aerodynamic characteristics of separated-box girders are difficult to analyze due to the complex outline and flow interference between upstream and downstream box girders.

The mechanism of VIV has been studied for over a hundred years and achievements are numerous, especially during the last decades. Fundamental characteristics of single circular cylinder VIV have been summarized by Bearman (1984), Griffin (1995), and Williamson and Govardhan (2004). According to these sources, vibration is mainly caused by the interaction between structures and wake flow or shear layer, which is determined by the Reynolds number and structural properties. As a function of inflow velocity, the amplitude of VIV can be divided into two or three branches with different mass ratios, $m^* = m/m_d$, where m is the mass of cylinder and m_d is the mass of the fluid displaced by cylinder (Govardhan and Williamson, 2000). There are an initial branch and a lower branch for high mass ratio cylinders ($m^* > 10$), and an initial branch, an upper branch and a lower branch for low mass ratio cylinders ($m^* \leq 10$). In addition to the amplitude, the vortex-shedding mode may also vary on the different branches. For the low mass ratio cylinders, two single vortices form per vibrational cycle on the initial branch, named the 2S mode by Williamson and

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Roshko (1988). On the upper branch, two pairs of vortices form per cycle; in each pair the following vortex is much smaller and weaker than the leading vortex, so this mode is designated the $2P_O$ mode (Morse and Williamson, 2009). As the reduced velocity increases, the vortex-shedding mode changes to the $2P$ mode on the lower branch, in which the two vortices in a pair are equally strong.

‘Lock-in’ is a classic scenario of VIV, which is defined as the matching of the frequency of the periodic wake vortex shedding and the oscillation frequency of the body (Khalak and Williamson, 1999). Cylinders with a mass ratio over 100 were firstly studied by Bishop and Hassan (1964) and Feng (1968) in wind tunnels. The vortex-shedding frequency, f_{vortex} , remains near the natural frequency, f_n , in the lock-in region, and does not increase with the reduced velocity U^* . For lower mass ratio ($m^* \sim O(10)$) cylinders, the ratio $f^* = f_{vortex}/f_n$ does not have to be precisely unity due to the effect of added mass. As noted by Bearman (1984), f_{vortex} is higher than f_n in the lock-in region, and increases as the mass ratio m^* becomes smaller (Khalak and Williamson, 1997). A recent work by Mittal (2016) studied the lock-in phenomenon more meticulously with a numerical method. According to the research, lock-in occurs when the energy ratio, E_r , of the unstable eigenmodes of the coupled fluid-structure system is above a certain threshold. The modes of lock-in can also be classified as fluid and elastic modes, for the high mass ratio cylinders, and fluid-elastic modes, for the low mass ratio cylinders, according to the coupling level of the eigenmodes.

When there are two or more structures in an array, the issue becomes more complex due to flow interference between upstream and downstream bodies, such as riser pipes on offshore drilling platforms and overhead power lines. The wake of the upstream body may impact the downstream body and induce vibrations, known as wake-induced vibration (WIV). Research shows that the shear layer in the gap begins to roll up when the separation distance extends beyond a critical x_0 , and this rolling up is believed to be the factor inducing WIV. Zdravkovich (1985) identified flow interference regimes through experiments with two stationary cylinders in different relative locations. Igarashi (1981) classified the flow interference of tandem cylinders into six stages according to the separation distance. Assi et al. (2010) conducted experiments on a tandem system, in which an elastically mounted cylinder followed a static one, and results showed that the motion of the downstream cylinder could be controlled by the wake of the upstream cylinder. When two elastic cylinders are independently mounted, the peak amplitude of the downstream cylinder can be approximately 50% higher than that of an isolated cylinder, according to Assi et al. (2006). Bearman (2011) investigated the lift force characteristics of the downstream cylinder and found that WIV is also a form of VIV. Sometimes there are structural connections between the separated bodies, such as bundle conductor spacers. Therefore, the vibrations of upstream and downstream bodies are strongly coupled and the flow state in the gap will also be affected by the synergistic oscillation of the bodies, yet studies on dependently and elastically mounted tandem systems are relatively lacking.

Among all the flow-induced vibration researches, most of them are concentrated on streamline structures or circular cylinders, while less attention has been paid to the VIV characteristics of polygonal bluff bodies. Taking a rectangular section cylinder for example (Nakamura and Nakashima, 1986), the vortex formed at the leading edge transmits along the structures surface and dissipates or sheds at the trailing corners; meanwhile, the flow separates at these trailing corners and forms a vortex. VIV commences when the leading vortex coalesces with the wake vortex due to boundary layer instability, known as ‘low-speed vortex excitation’. Bentley and Mudd (2003) discussed the vortex-shedding stages of single and dual bluff bodies, showing that the flow interference between leading and trailing structures was more complex than that of tandem circular cylinder systems, due to the different downstream body shapes.

Research on VIV of bridges arose after the collapse of the Tacoma Narrows Bridge in 1940. As a compromise between feasibility for construction and aerodynamic stability, box girder sections are usually designed as flat polygons, which have fixed separation points but low-level flow separation. VIV stability of the separated-box girder, a recently invented girder style, has been widely investigated in recent years. Field monitoring data of a built SBG bridge was analyzed by Li et al., (2011, 2014b), and the vertical VIVs with different vibration modes were observed over a wind velocity range of 6–13 m/s. Laima et al. (2013) and Li et al. (2014a) experimentally investigated VIV characteristics of a separated-box girder without and with control. It was concluded that the dynamic response of bridges is highly influenced by flow interference in the central gap, and the gap width is a dominant factor in the occurrence of VIV. Chen et al. (2014) performed particle image velocimetry (PIV) measurements with a stationary SBG model, probing the relationship between gap widths and trigger conditions for VIV. The results show that VIVs are induced only if the gap is wide enough to encompass two vortices. If the gap is too wide, VIV will be suppressed. Although VIVs of the SBG bridge have been comprehensively investigated through both wind tunnel testing and field monitoring, the trigger mechanism for VIV of the SBG model, the flow field around the dynamic SBG model and the interaction between VIV and the flow field are not clearly understood and need further study.

To study the VIV mechanism of separated-box girders more deeply, more refined experiments were conducted in this research and some new phenomena were observed and discussed. The dynamic responses of a SBG model show that VIVs are repeatedly induced in different wind velocity intervals, in which the vibrational frequencies are all locked to the natural frequencies in vertical and torsional modes. The present study has managed to synchronize the instantaneous vortex fields and oscillation phase, and visualize the interaction between flow and structure. WIV is found to happen with the dynamic model and there is a mutual promotion between WIV and VIV. The vortex-shedding mode changes from the $2S$ mode to the $2(2S)$ mode. A special lock-in occurs in which the model motion synchronizes with the vortex-shedding at a ratio of 1:2.

The experimental details, including parameters, arrangement of the model and test conditions are presented in §2. The measurements of dynamic response, vortex-shedding frequency and flow fields are given in §3, including some discussion. Conclusions of the investigation are summarized in §4.

Table 1
Physical parameters of the model.

Parameters		Unit	Value
Length	L	m	0.4800
Width	B	m	0.1800
Depth	D	m	0.0175
Equivalent mass	m	kg/m	1.5490
Mass moment of inertia	I_m	kg m	0.0024

were measured by conducting a free decay test in quiescent air. The results were highly repeatable and are shown in Table 2. The mass and damping ratio, $m^*\zeta$, is 0.63 for the dynamic model in the vertical direction.

Dynamical response was measured by two laser displacement sensors and the measurement locations were symmetrically arranged about the medial axis (shown in Fig. 2(b)). The vertical and torsional motion is defined by the following equation:

$$y = (y_l + y_t)/2 \quad (1)$$

$$\theta = 180 \times (y_l - y_t)/\pi d \quad (2)$$

where y is the vertical displacement, θ is the torsional angle, d is the distance between the two measurement locations, and y_l and y_t are the vertical displacements of the leading and trailing measurement locations, which are labelled in Fig. 2(b). The accuracy of the twist angle calculated by this equation is sufficient, since the amplitude of torsional vibration is relatively small.

The inflow wind velocities were in the range of 0.5–3.1 m/s and the corresponding Reynolds numbers, following Eq. (3), were in the range of 587–3,641, when taking the depth of model as the characteristic dimension.

$$Re = UD/\nu \quad (3)$$

where U is the inflow wind velocity, D is the characteristic dimension and ν is the kinetic viscosity of air.

2.3. Measurement apparatus

Dynamical response was measured by laser displacement sensors (Keyence LK-G3000, Japan). The accuracy of the sensors was within 0.02% of the 200 mm available linear range, giving a precision of $0.002D$. Sensors were reset to zero before the tests. A hot wire probe (Dantec Dynamics, Denmark) was used to determine the vortex-shedding frequency (VSF) around the model. The sampling frequency of both dynamical measurements and VSF measurements was 1000 Hz, and the sampling time was over 120 s at every test velocity when the motion state became stable. Flow fields around the model were measured by a PIV system at VIV-sensitive velocities. The section for PIV measurements was located at 1/3 span (see Fig. 2(b)), so that the flow fields in the central

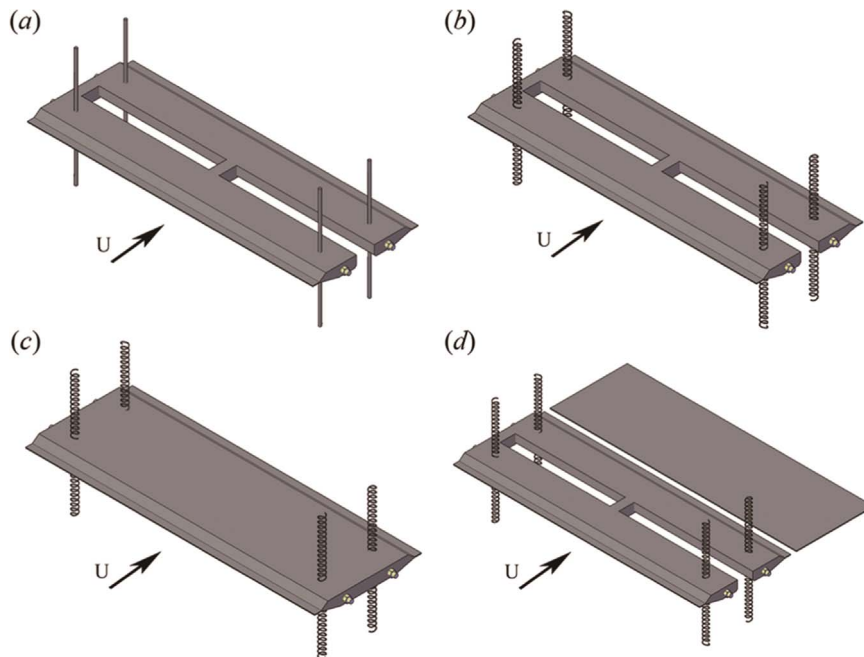


Fig. 3. Sketches of the testing models. (a) The static model, (b) the dynamic model, (c) the gap-covered model and (d) the wake-split model.

Table 2

Dynamical parameters of the test models.

Parameters		Dynamic model	Gap-covered model	Wake-split model
Vertical frequency	f_v (Hz)	6.77	6.70	6.77
Torsional frequency	f_t (Hz)	8.10	8.10	8.10
Vertical damping ratio	ζ_v (%)	0.7	1.0	0.9
Torsional damping ratio	ζ_t (%)	3.8	3.8	3.9

gap could be captured by the camera. The flow was seeded with oil droplets having a normal diameter of 1–5 μm . Two Nd: YAG pulse lasers (Beamtech Vlite-200, China) were used to provide illumination with 1.0 mm-thick laser sheets. The laser was divided using a beam splitter and emitted from two guiding beam arms to illuminate both the upper and lower side of model at the same section (Fig. 1). Imaging was performed using a PCO 1600 (PCO, Germany) camera with a resolution of 1600×1200. The PIV system was timed by a digital delay generator (Berkeley Nucleonics 575–8 C, US), which triggered the laser pulse and image acquisition. Since the oscillations were highly periodic and sinusoidal in all VIV-sensitive velocity intervals, the PIV measurements were sampled at eight phases in one cycle for phase-locked averaging. Over 200 image pairs for each phase were recorded. The filming domain was 264 mm in length and 198 mm in width, in which 605 pixels were equal to 100 mm in reality. The image data were processed by using a frame-to-frame cross-correlation technique with interrogation windows of 24×24 pixels² and 75% effective overlap. After processing, wind velocity vector fields were obtained with a size of 263×197. Each vector represented the average velocity of an area of 3.97×3.97 mm², and the distance between adjacent vectors was 1 mm.

3. Results and discussion

3.1. Dynamical response

The dynamical responses of three types of model (i.e., dynamic model, gap-covered model and wake-split model) were measured. The amplitudes versus reduced wind velocities are shown in Fig. 4. For the dynamic model, there are four separate wind velocity intervals, in which vertical vibration and torsional vibration are induced repeatedly. As the wind velocity increases, vertical motion is induced first with an amplitude of 0.086D, accompanied by a slight torsional vibration. A single-degree torsional motion is induced subsequently in a narrow velocity interval. Next, vertical vibration is observed again, but with a low amplitude. Last, the second torsional vibration occurs with the same torsional amplitude as before but over a wider wind velocity interval. For convenience, the four intervals will be referred to as V1, T1, V2 and T2 hereafter, according to wind velocity from low to high. Although the vibrational amplitude is not as dramatic as the VIV of circular cylinders, it is considerably high for bridges with such a streamlined profile and high mass ratio. For the wake-split model, there is only one VIV velocity interval, which is the same as V1 with an amplitude that is approximately 20% lower than that of the dynamic model, implying that the wake vortex shedding contributes less than the gap vortex shedding in this condition. No significant vibration peaks are observed for the gap-covered model, which means that VIV cannot occur without the gap.

The heat maps of power spectral density (PSD) for the dynamic model and the wake-split model are shown in Fig. 5, which presents the vibrating frequencies at different velocities. For the dynamic model and the wake-split model, the frequency is locked to

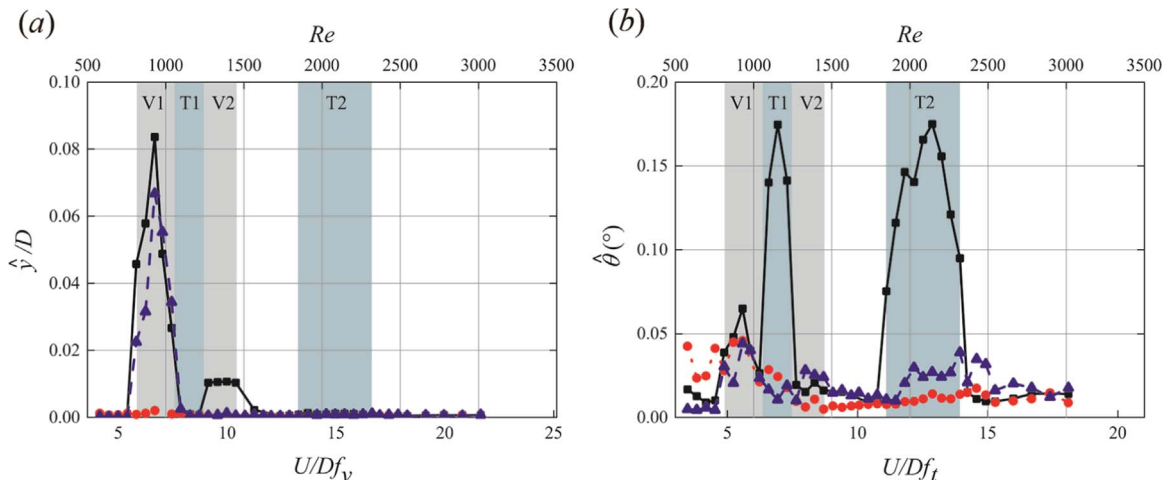


Fig. 4. Dynamical responses of the testing models: (a) vertical and (b) torsional. The black squares represent the dynamic model, the blue triangles represent the wake-split model and the red circles represent the gap-covered model. $\hat{y}$ is the RMS vertical displacement, $\hat{\theta}$ is the RMS torsional angle, D is the depth of the model and U is the inflow wind velocity. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

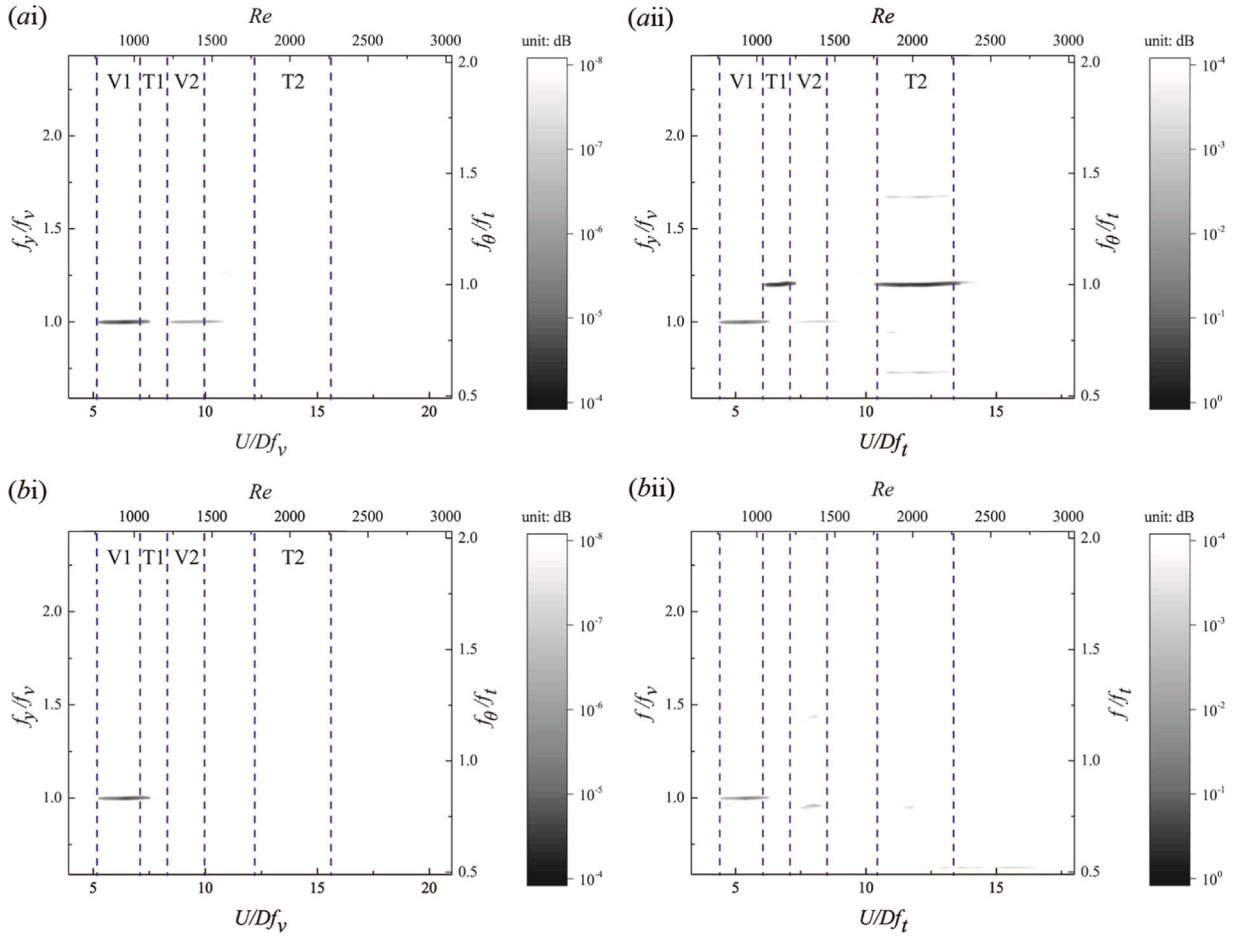


Fig. 5. Heat maps of the power spectral densities for (a) the dynamic model and (b) the wake-split model in (i) vertical and (ii) torsional modes. f_y is the vertical vibrating frequency and f_θ is the torsional vibrating frequency.

the natural frequency in the corresponding direction (vertical and torsional) in the VIV wind velocity intervals. There is no dominant frequency when VIV does not occur because the vortex is very weak due to the streamlined profile and high mass ratio, therefore the vortex-induced force is not strong enough to excite the model. VIV does not occur on the gap-covered model and the PSD heat map (which is omitted here) is blank.

Segments of the response time history and characteristics of the vibrational frequency at the velocity with the highest amplitude in each interval are shown in Fig. 6 for the dynamic model, in which f represents the vibrational frequency. The oscillations in the corresponding directions are highly periodic and harmonic in all four of the wind velocity intervals. Both vertical and torsional motions exist in V1, and the frequencies are equal to the natural frequency of vertical vibration, implying that the coupled vibration is dominated by the vertical vibration. There is a constant phase difference between the vertical and torsional vibrations of approximately 120° . Distributions of the power spectrum density are concentrated and the vibrational frequency is equal to the natural frequency of the model in the corresponding vibration mode, which is a typical phenomenon of 'lock-in'. The oscillation appears to be a type of vortex-induced vibration.

Comparing the responses of these three types of test model, it can be concluded that the central gap is a crucial factor in inducing all four types of VIVs because all of the VIVs are suppressed when the gap is covered. This is mainly due to the streamlined profile of the model without a gap. The vortex may be generated or enhanced within the central gap. The wake splitter plate, which prevents the alternating shedding of the wake vortex, inhibits VIV in the three wind velocity intervals other than V1, meaning that the vibration occurring in V1 is mainly induced by the gap, and the other three VIVs are jointly controlled by gap and wake vortex-shedding.

3.2. Flow field characteristics around the static model

To have a clear understanding of the VIV trigger mechanism, PIV measurements were conducted at VIV-sensitive velocities. Some key locations on the model where vortices may form are specified in Fig. 7 for clarity. The time-averaged velocity fields in the horizontal and vertical directions are shown in Fig. 8. Generally speaking, the velocity fields in the four intervals have similar

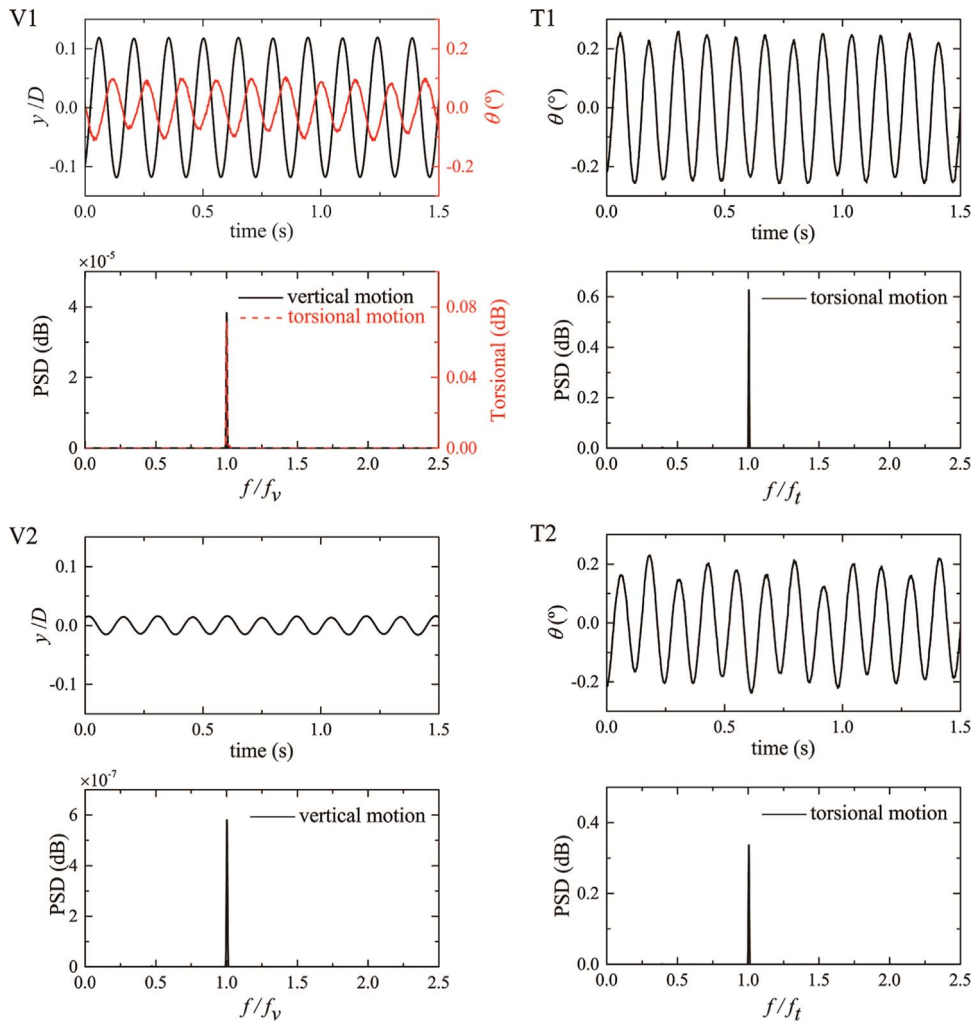


Fig. 6. Time histories and the power spectral density of the vertical and torsional motions. Upward motions and pitching-up motions are taken to be positive. The exact inflow velocities are 0.66 m/s, 0.95 m/s, 1.14 m/s and 1.80 m/s for V1, T1, V2 and T2, respectively.

features, which are mainly determined by the outline of the test model. The maximum horizontal velocity is located at the top and bottom of the leading wind fairing, while the velocity is close to or even less than zero in the central gap and behind the trailing wind fairing. The shear layer near the model gets thinner with increasing velocity. The vertical velocity is much lower than the horizontal velocity due to the streamlined profile. Maximum and minimum vertical velocities appear around the leading and trailing wind fairings, and the vertical velocities in the gap are close to zero.

Turbulence kinetic energy distributions of the flow fields are used to assess the variability of the flow stream. T.K.E. is defined as the root-mean-square k , following the equation $k=(u^2+v^2)/(2U_\infty^2)$, in which u and v are the wind velocities in the horizontal and vertical directions, respectively, and U_∞ is the inflow wind velocity. If the T.K.E. is close to zero, it means that the flow is basically standing at that position and cannot fluctuate the model, thus it is not inducing vibration. T.K.E. distributions of the static model are shown in Fig. 9, from which we can see that the T.K.E. spreads only in the wake zone of the model. With increasing wind velocity, the distribution range becomes wider and closer to the model. The maximum T.K.E. in V1 is 0.05 and rises to 0.07 in T1; however, it drops to 0.06 in V2 and 0.04 in T2. The flow field in the central gap is quite stable.

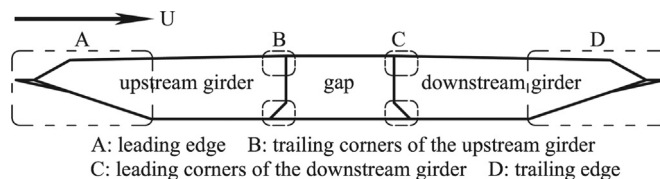


Fig. 7. A sketch of some key locations on the bridge girder.

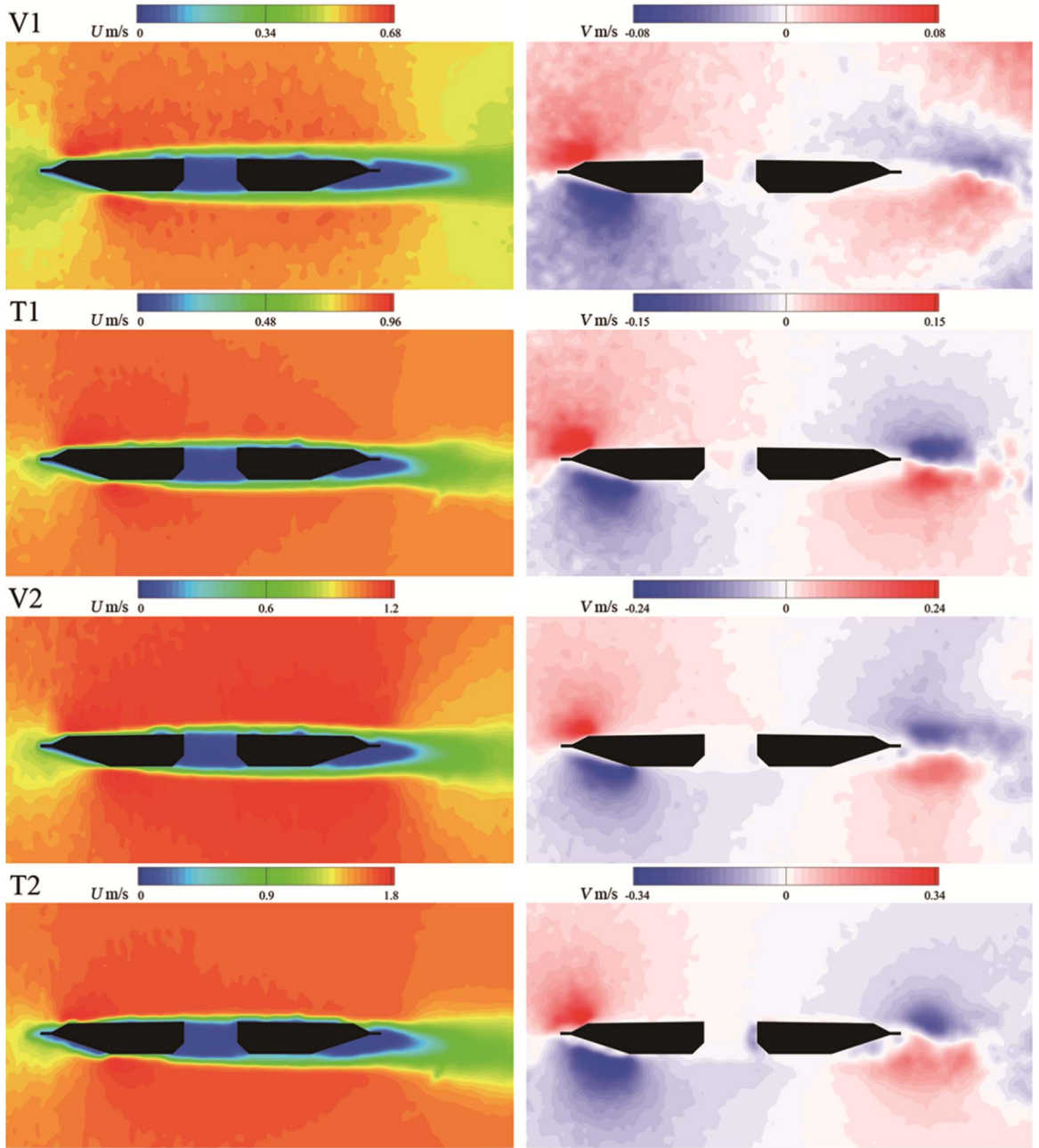


Fig. 8. Time-averaged velocity fields around the static model. U is the horizontal velocity and V is the vertical velocity, taking streamwise and upward as positive, respectively. The corresponding inflow velocities are 0.66 m/s, 0.95 m/s, 1.14 m/s and 1.80 m/s for V1, T1, V2 and T2.

For near-wall turbulent flow, a vortex may not be distinguishable from the shear layer by its vorticity value (Chen et al., 2014). Therefore, the swirling strength is chosen to visualize the vortex around the model in this section. Swirling strength is defined by Zhou et al. (1999) as the imaginary part of the complex eigenvalue of the following equation:

$$D_{2-D} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{bmatrix} \quad (4)$$

where D_{2-D} is the two-dimensional flow velocity gradient tensor, u is the horizontal velocity, v is the vertical velocity, and x and y are the horizontal and vertical coordinates, respectively.

The swirling strength distributions around the static model are presented in Fig. 10. Four phases in one cycle are included to describe the evolution of the vortex fields. In this study, the swirling strength of a clockwise vortex is defined to be a negative value and shown in blue,

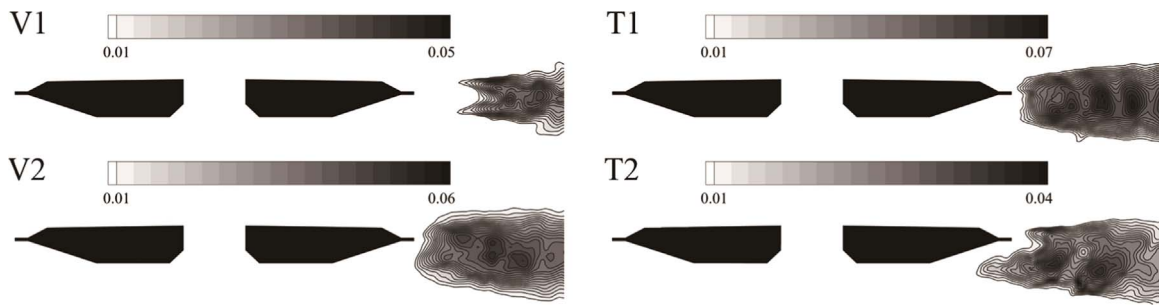


Fig. 9. Turbulence kinetic energy distributions around the static model. The corresponding inflow velocities are 0.66 m/s, 0.95 m/s, 1.14 m/s and 1.80 m/s for V1, T1, V2 and T2.

while that of an anticlockwise vortex is a positive value and red. The streamlines are traced out in the background. It can be observed that periodic vortex shedding only exists in the wake region for all the four wind intervals, which means that the fluctuation of the wind velocity is strong in the wake, therefore the T.K.E. value is large in the wake, as shown in Fig. 9. The swirling strength of the wake vortex gets stronger with increasing velocity, and the scale also gets larger. The strengths of the clockwise and anticlockwise vortices are almost equal. The clockwise vortex always forms at the trailing wind fairing, while the location of the anticlockwise vortex formation moves forward from the wind fairing to the middle lower surface of the downstream box girder as the wind speed increases. Note that no significant separation occurs at the leading wind fairing. Therefore, we do not think that a vortex exists at the leading edge, even though the swirling strength value there is persistently high. The flow interference in the central gap is presented by streamlines because the velocity is so low that the swirling strength is close to zero. There are two reverse vortices in the gap in V1; the upper one is clockwise and larger in scale, and the lower one is anticlockwise and smaller. The inequality of the two vortices is mainly due to the chamfering of the lower corners. In T1 and V2, the clockwise vortex grows bigger in scale, while the anticlockwise one gets compressed into the left bottom of the gap. In T2, the gap is filled by the clockwise vortex and the anticlockwise vortex is hardly visible. For the static model under all four wind intervals, the vortices within the gap are standing—there is no alternating vortex shedding. The flow interference between upstream and downstream bodies is similar to stage C as defined by Igarashi (1981), such that wake-induced vibration cannot occur in this situation.

The similarities in time-averaged velocity fields, T.K.E. distributions and vortex fields of the four wind intervals present a slight Reynolds effect over the tested Re range.

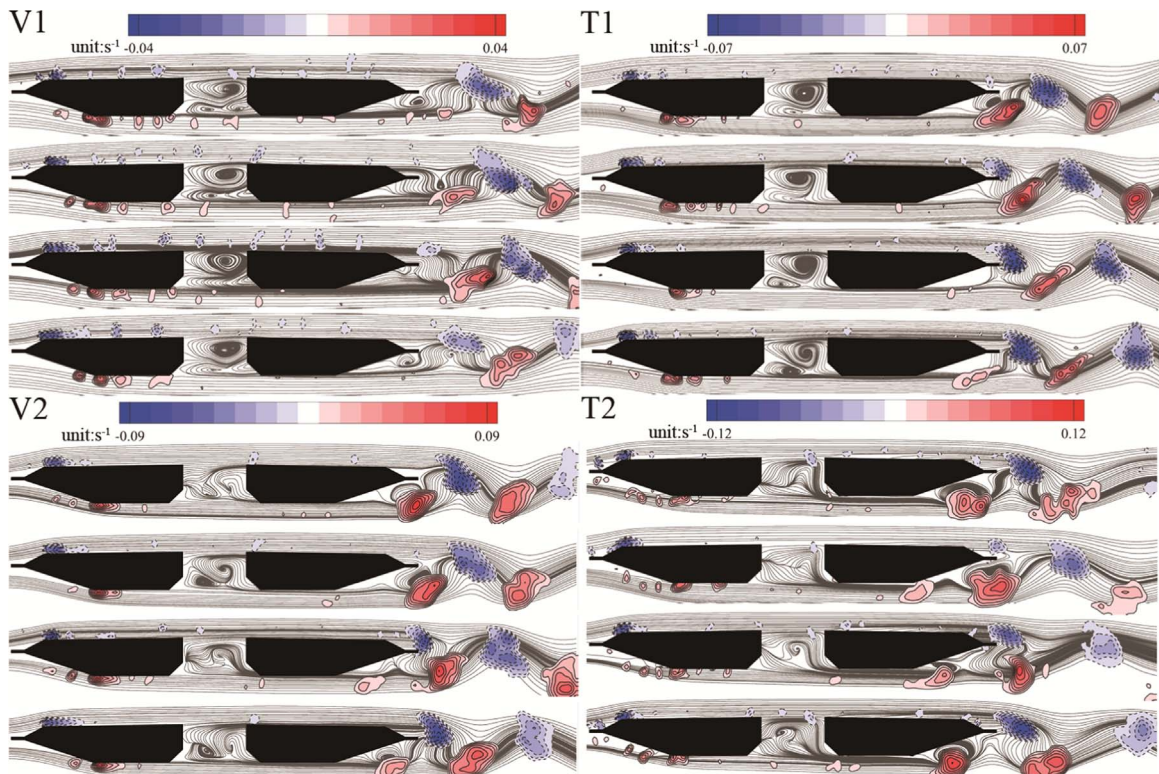


Fig. 10. Evolution of the flow fields around the static model. The corresponding inflow velocities are 0.66 m/s, 0.95 m/s, 1.14 m/s and 1.80 m/s for V1, T1, V2 and T2.

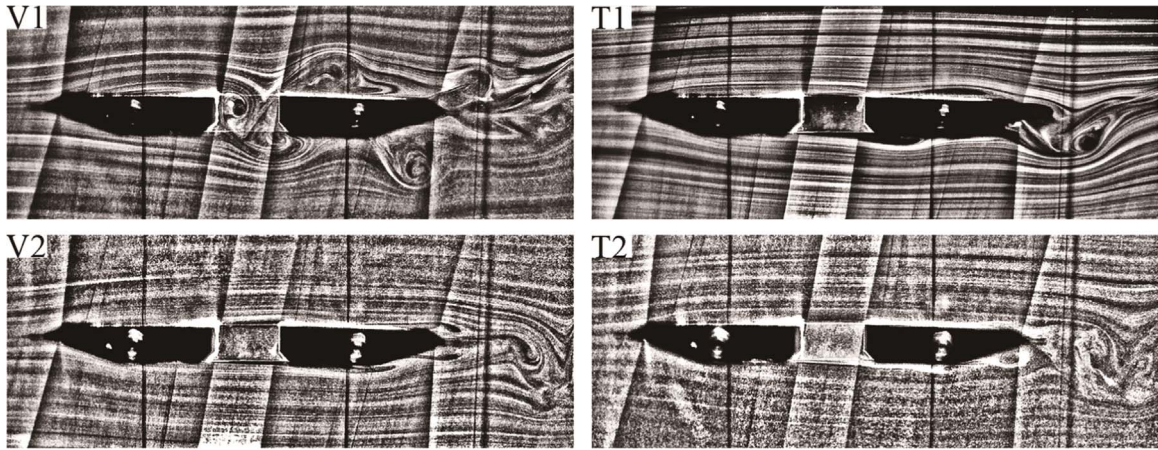


Fig. 11. Instantaneous flow fields around the dynamic model. The corresponding inflow velocities are 0.66 m/s, 0.95 m/s, 1.14 m/s and 1.80 m/s for V1, T1, V2 and T2.

3.3. Flow field characteristics of the dynamic model

Instantaneous flow fields of the vibrating model are visualized by the smoke streamline photos in Fig. 11, from which the vortex and streamlines can be directly observed. It can be seen that there is no significant flow separation at the leading edge of the upstream body, and a vortex may appear at the central gap, upper and lower surfaces of the downstream body and wake zone, depending on the wind velocity. Note that the flow field in V1 significantly differs from that in the other VIV intervals. For example, the scale of the vortex is much larger and a vortex shed into the central gap is then transmitted along the downstream body and eventually shed into the wake.

The scale of the vortex around the surface of the downstream body is very small in T1, V2 and T2, and the vortex is not shed into the central gap, which makes the gap into a dead water zone. Moreover, the wake-shedding mode in V1 and T1 seems to be different from that in V2 and T2. More specific analyses will be carried out by velocity field calculation and post processing below.

The T.K.E. distributions of the dynamic model are shown in Fig. 12. The level of flow-structure interaction is revealed by comparing the results for the static model with those for the dynamic model. From the figures we can see that the T.K.E. in the wake zone is still widely spread in all the four wind intervals. The maximum T.K.E. in V1 is approximately 0.03, increasing to 0.06 in T1 and V2, but reducing to 0.04 in T2. The T.K.E. distribution area in the wake has a similar trend: small in V1, bigger in T1 and V2, and shrinking in T2. Compared with the distributions in the static model, the T.K.E. is lower around the dynamic model in V1 and T1, and the distribution area is wider. Additionally, a visible difference is the energy spreading into the gap zone and around the surface of the downstream girder in V1. The results agree with the visualized flow fields in Fig. 11. The fluctuation of the flow in the gap depends on the vibration of the model or the interaction between fluid and structures. Considering that the flow interference in the central gap is a crucial factor for wake-induced vibration, it is reasonable to say that WIV cannot occur in the static condition, it must be induced by other types of oscillation or perturbation.

The swirling strength distributions of the dynamic model are shown in Fig. 13–16, according to wind velocity intervals. To analyze the flow fields combined with model motion, 8 phases are included in one cycle, as labelled in the left of the figures. For all intervals, phase (i) and phase (v) represent the equilibrium position, while phase (iii) represents the highest position for V1 and V2, or the largest pitching-up angle for T1 and T2, and phase (vii) is the reverse. Since the flow interference in the central gap is crucial to the VIV and WIV as mentioned before, the streamlines in the central gap at each phase are also illustrated in Figs. 13–16.

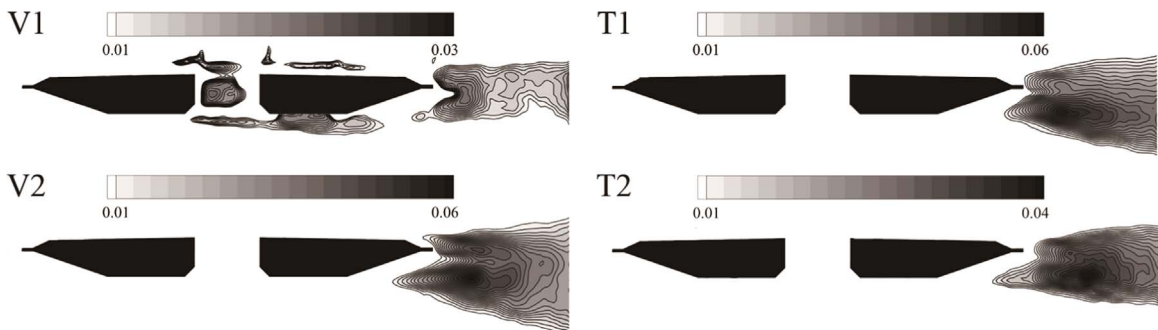


Fig. 12. Turbulence kinetic energy distributions around the dynamic model. The corresponding inflow velocities are 0.66 m/s, 0.95 m/s, 1.14 m/s and 1.80 m/s for V1, T1, V2 and T2.

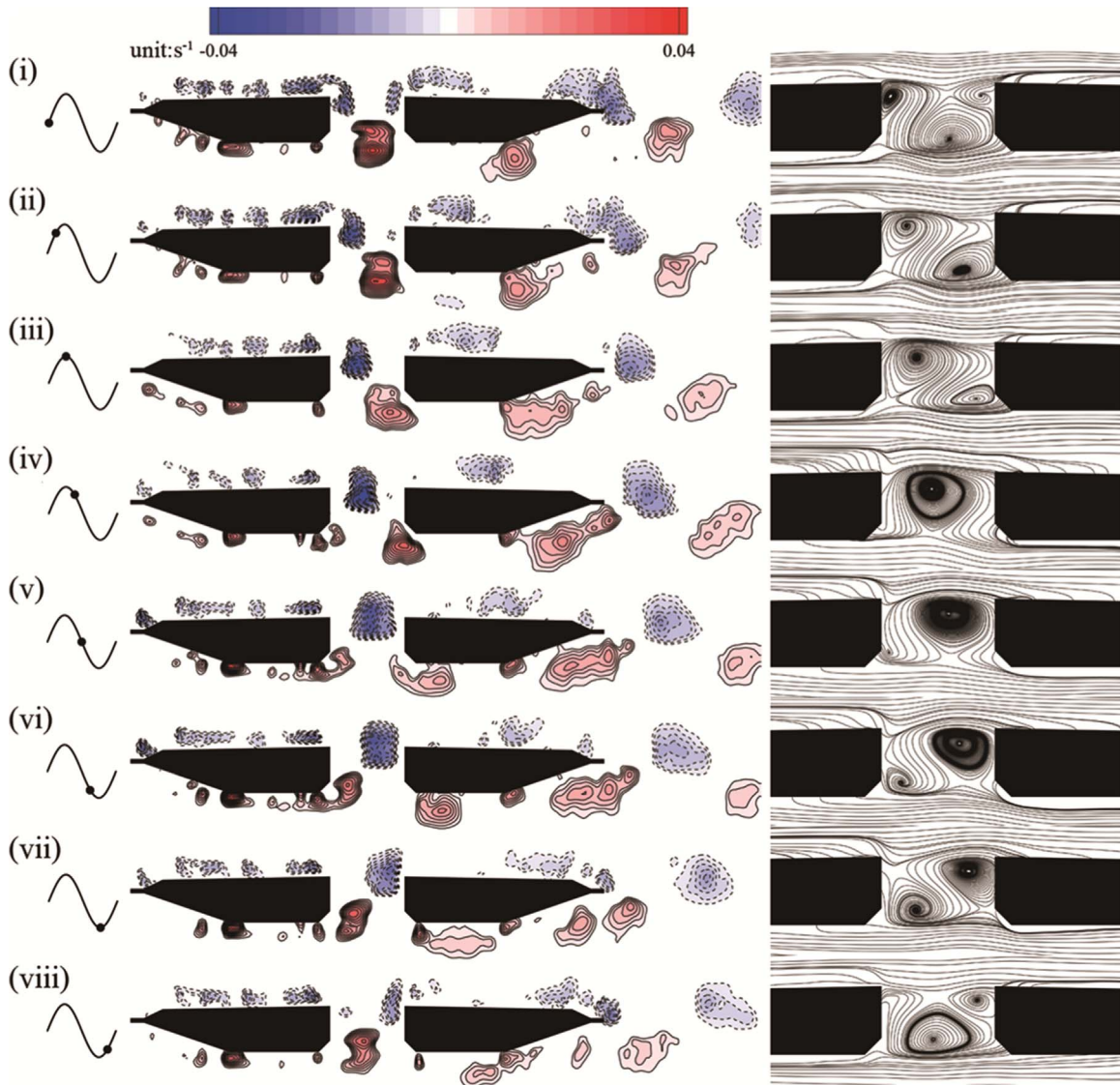


Fig. 13. Evolution of the flow fields around the dynamic model in V1. The inflow velocity is 0.66 m/s.

The swirling strength distributions of the dynamic model in the V1 interval are shown in Fig. 13. The results coincide well with the flow fields in Fig. 11, except for the vortex around the upstream box girder. The vortices around the upstream box girder are misidentified as vortex by the swirling strength algorithm. A pair of vortices form at the trailing corners of the upstream box girder and are periodically shed into the gap in one cycle, which is a classic Kármán vortex street in the 2S vortex-shedding mode, as defined by Williamson and Roshko (1988). The vortex continues downwind along the surface of the downstream box girder and eventually sheds again at the trailing wind fairing. The wake shedding shares a mode with gap shedding. The strength of the vortex reaches its maximum at the central gap and weakens gradually in the wake, implying that energy is exchanged between vortex and model motion. Taking the motion phase into account, the vortex lasts approximately 2.5 circles of vibration before it is completely shed in the wake. Using the anticlockwise vortex as an example, the vortex forms in phases (iii) to (vii) at the lower side of the central gap in the first circle, accompanied by the downward motion of the model, and reaches the downstream box girder at phase (iii) in the second circle. Then, the vortex is transmitted along the lower surface of the downstream box girder. Wake shedding occurs in phases (iii) to (vii) of the third circle. The clockwise vortex on the upper side behaves similarly, with a phase difference of a half circle. It is noteworthy that the vortex is shed synchronously in the wake and in the gap, which provides excitations to the model at two positions. The gap shedding is stronger than the wake shedding. By comparing the dynamical responses of the dynamic model and the wake-split model, as shown in Fig. 4(a), it can be concluded that wake shedding contributes less than gap shedding to the oscillation amplitude, since the amplitude only decreases by 20% when the wake-vortex shedding is suppressed.

The streamlines in the central gap, shown in Fig. 13, demonstrate the flow interference more precisely. It can be observed that the shear layer on the upper side rolls down in phase (viii) to phase (iii) in the next cycle, while the shear layer on the lower side rolls

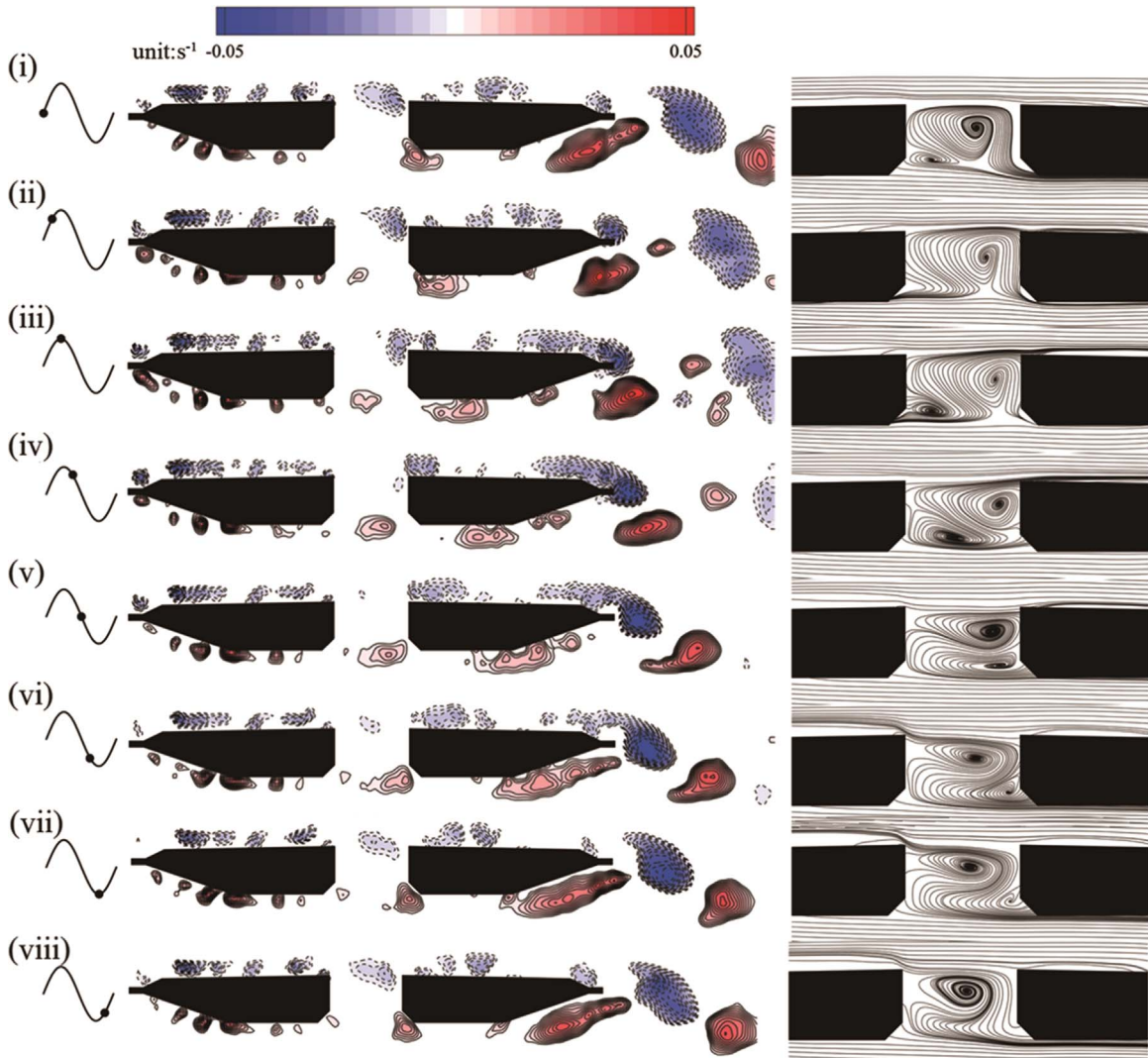


Fig. 14. Evolution of the flow fields around the dynamic model in T1. The inflow velocity is 0.95 m/s.

up in phases (iv) to (vii). The periodic rolling of the shear layer is a signal of wake-induced vibration.

Flow fields of the dynamic model in the T1 interval are presented in Fig. 14. The Kármán vortex street still exists in the gap, but the vortices are not fully strengthened as they are in V1. Both the scale and strength of the vortex is undeveloped. In the gap, the anticlockwise vortex on the lower side commences at phase (ii) and the clockwise vortex on the upper side forms at phase (vi). Then, the vortex touches the downstream box and is transmitted to the wake along the girder surface. The vortex is greatly enhanced during the process of wake shedding, both in scale and strength. The vortex-shedding mode in the wake in T1 is also a $2S$ mode. From the streamlines in the central gap we can see that the large clockwise vortex persists, accompanied by periodic changes in shape and slight movement of the vortex core. The process of vortex shedding on the lower side is still clear, but the scale is much smaller than that in V1. The flow interference seems to be a critical state of stage D to stage E as proposed by Igarashi (1981), in that the rolling of the shear layer still exists but at a relatively low level.

Taking the results in Fig. 12 into consideration, the flow state in gap is basically stable in T1. The level of WIV is very low compared to that in V1. So, why is the interaction level in this case not as high as in V1? On one hand, the boundary of the gap is close to the rotational center, where the tangential velocity of the model is relatively low, so the oscillations of the model can hardly promote vortex generation in the central gap. On the other hand, the critical distance x_0 for rolling of the shear layer changes with wind velocity, so the shear layer in the gap cannot roll up and perturb the downstream girder as in V1. Therefore, the WIV in T1 is not as strong as in V1.

Flow fields around the model in V2 are shown in Fig. 15. The vortex mainly forms at the leading corners of the downstream box girder rather than in the gap. Note that two pairs of vortices shed in one oscillational cycle, which means that the vortex-shedding mode changes from the $2S$ mode to another mode. Specifically, a pair of vortices form at the leading corners of the downstream box girder and are transmitted downwind, after which they successively shed into the wake in phases (vii) and (viii), when the model

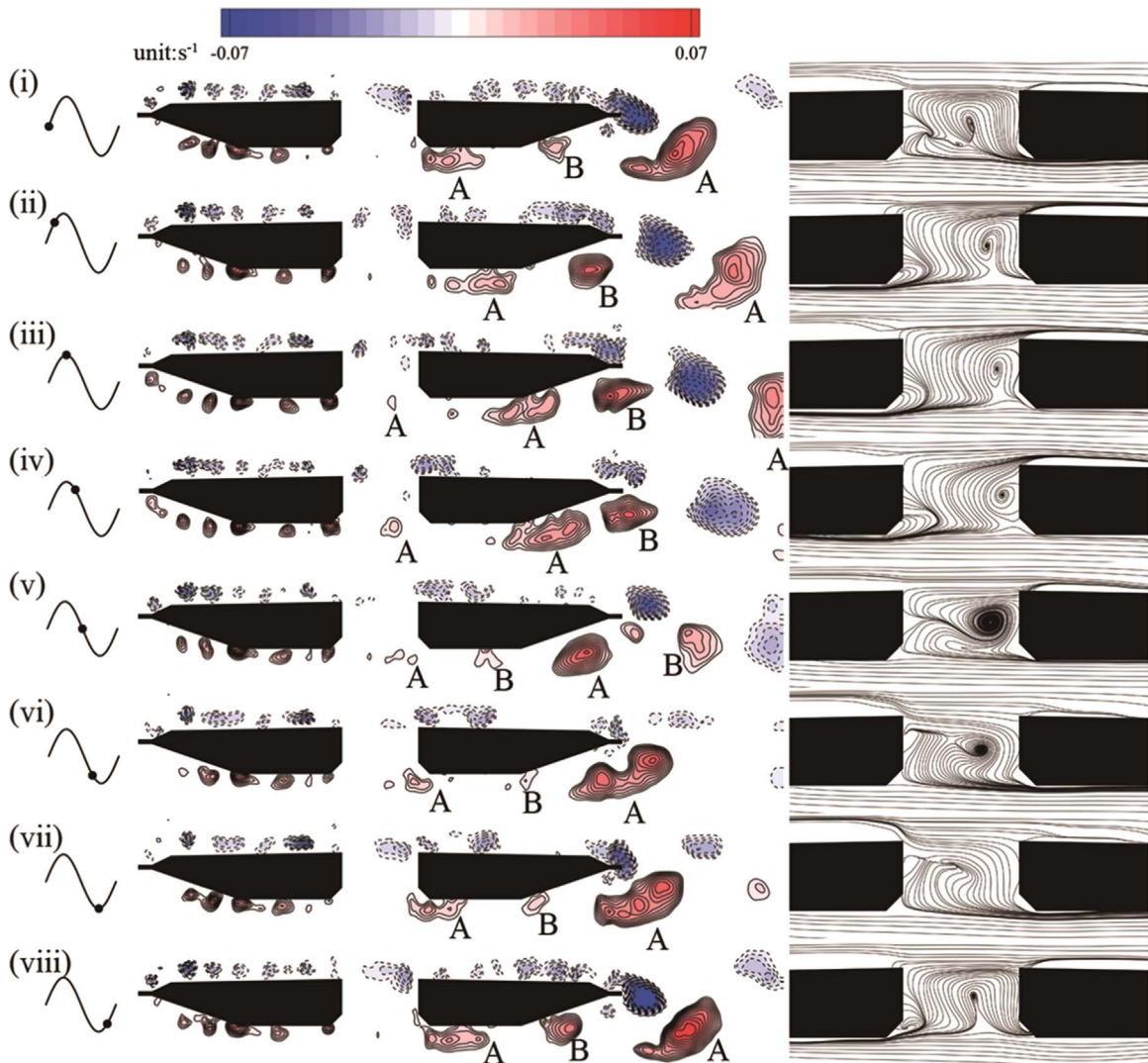


Fig. 15. Evolution of the flow fields around the dynamic model in V2. The inflow velocity is 1.14 m/s.

reaches its lowest location. The anticlockwise vortices of this pair are labelled A in Fig. 15 to show the trace. Moreover, another pair of vortices form at the middle of the downstream box girder and shed simultaneously into the wake in phases (iii) and (iv), when the model reaches its highest location. Yet this pair does not last long before it dissipates in the wake at phase (vi). The trace of this pair is labelled B in the figure. The streamlines in the central gap show that there is no visible vortex shedding or rolling up of the shear layer as in V1 and T1. The WIV does not occur anymore in this Reynolds number range because there is no periodic roll up of the shear layers in the gap.

According to the definition of the $2P$ mode by Williamson and Roshko (1988), two pairs of opposite-signed vortices alternately shed in one vibrational cycle, but the shedding order of the two vortices in each pair are different. However, the anticlockwise vortex sheds before the clockwise vortex in both of the 2 pairs in V2, which means that the present vortex-shedding mode is not a $2P$ mode. Following the naming rule of Williamson and Roshko (1988), this vortex-shedding mode could be designated $2(2S)$. The same mode was also found by Zhao et al. (2014) for a low mass-damping ratio ($m^*\zeta=5.83$) square cylinder at $U^*=8$ with an attack angle of 20° . Two different pairs of vortices shed in one vibrational cycle, where the vortices are fed by the vorticity produced at rear and sides due to the body motion. Based on the location of the vortex formation in the present study, the two pairs of vortices are very likely to be fed by the vorticity produced around the gap, including the trailing part of the upstream box and the leading part of the downstream box.

Vortex fields in T2 of the dynamic model are illustrated in Fig. 16. The vortex forms at the top and bottom surfaces of the downstream girder, and swirling strength becomes more intense with increasing wind velocity. There are also two pairs of vortices successively shed in one circle and the swirling strength of the two pairs of vortex is equal to each other, which implies that the vortex-shedding mode is a typical $2(2S)$. The flow state in the central gap is rather stable, such that the flow interference cannot perturb the model directly. Summarily, the flow fields are quite different in the four intervals. With the increase in velocity, vortex-

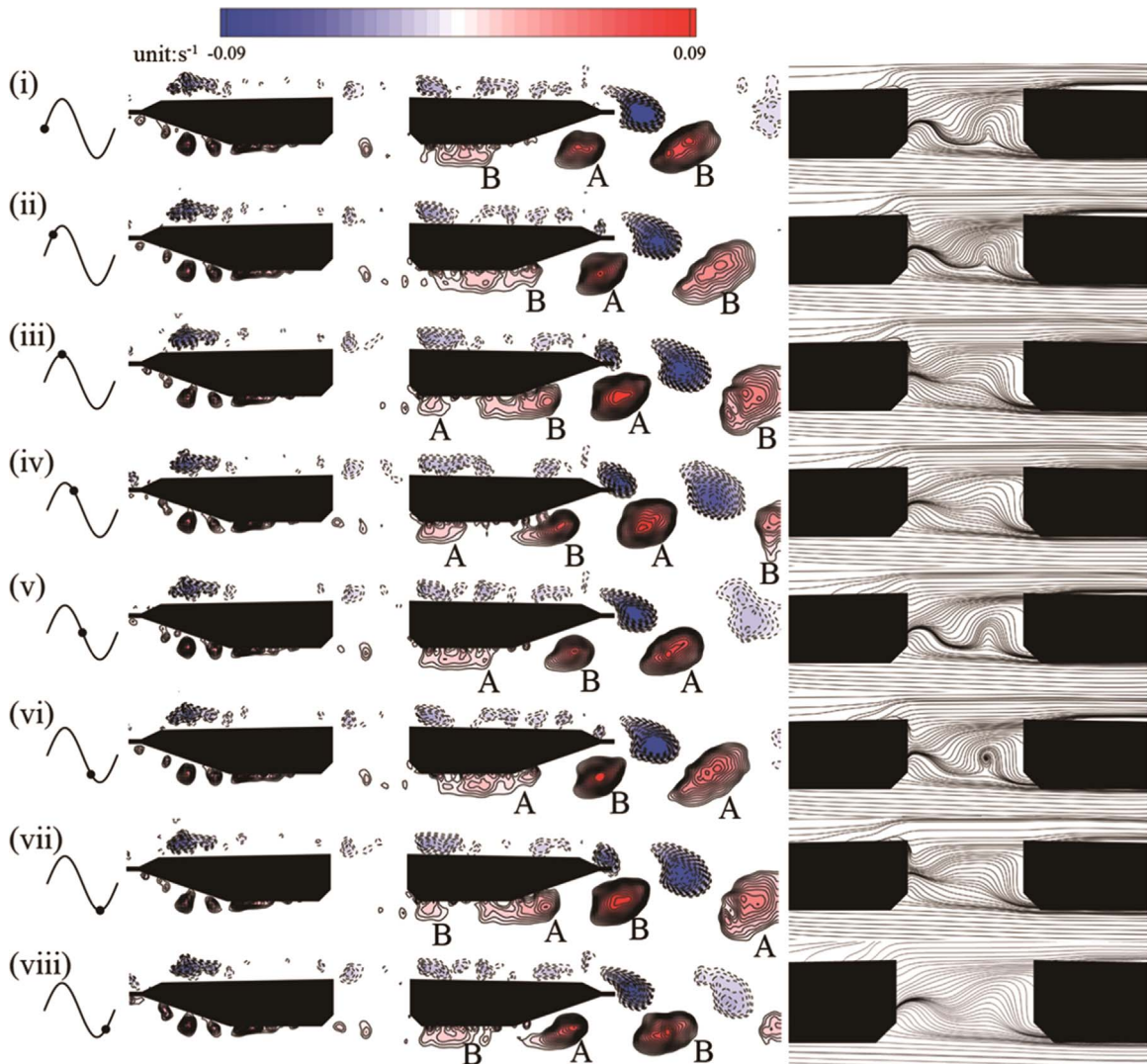


Fig. 16. Evolution of the flow fields around the dynamic model in T2. The inflow velocity is 1.80 m/s.

shedding mode changes from 2S mode to 2(2S) mode. Wake-induced vibration commences in V1 when the shear layer rolls up and the Kármán vortex street forms in the central gap. However, it does not arise in the other three intervals because the critical distance x_0 is a function of Reynolds number and the shear layer cannot roll up into the gap at higher wind velocities. Additionally, the location of vortex formation moves from the trailing corners of the upstream body to the top and bottom surfaces of the downstream body as velocity increases. The occurrence of VIV in V1 and T1 is easy to understand, but the inducing factor of VIV in V2 and T2 seems to be confusing.

If we take the upstream and downstream box girders to be two individual structures, the inflow of the downstream body is disturbed and obstructed by the upstream body. Therefore, the downstream body is experiencing a very different wind condition from the upstream body at the same time. Movement characteristics of the vortex are worthy of consideration because they represent the virtual wind velocity at the downstream box girder to a certain extent. It can be observed from the figures that the velocity of the vortex is not equal to that of the free stream, nor is it constant at a given inflow velocity. Considering the condition in V1, the vortex moves a distance equal to the gap width along the wind in the first circle since the vortex is generated at the gap, and the velocity is approximately equal to $1.7Df_v$. In the second circle, the vortex moves a distance equal to the downstream box width, and the velocity is approximately $3.0Df_v$. In the third circle, in which the vortex sheds into the wake and goes away, the velocity is still $3.0Df_v$. Meanwhile, the inflow velocity is $5.8Df_v$, which is about twice of the velocity of vortex. The velocity of vortex movement in the wake zone is approximately $3.4Df_v$ in T1. Meanwhile, the inflow wind velocity is $6.9Df_v$, which is also twice the velocity of the vortex. The velocity of vortex A in V2 is approximately $5.31Df_v$, which is very close to the inflow velocity in V1. Meanwhile, the inflow velocity is $9.98Df_v$, which is also approximately twice the velocity of vortex movement. The velocity of vortex movement is $4.34Df_v$ in T2 with an inflow velocity of $12.70Df_v$. Rough calculations show the velocity of vortex movement is approximately half the free stream velocity. The results agree well with the research by Shiraishi and Matsumoto (1983) that the vortex velocity is approximately 0.6 times the inflow velocity.

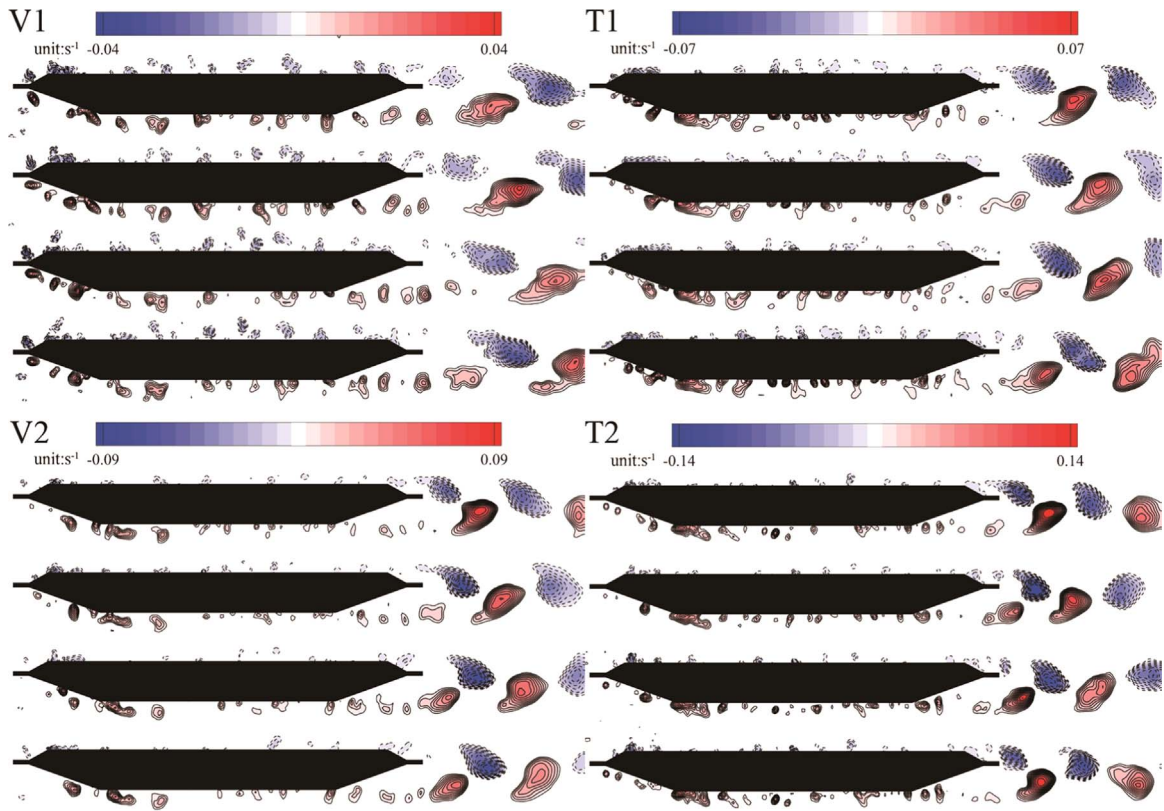


Fig. 17. Evolution of the flow fields around the gap-covered model in V1. The corresponding inflow velocities are 0.66 m/s, 0.95 m/s, 1.14 m/s and 1.80 m/s for V1, T1, V2 and T2.

To better understand the role of the central gap in triggering VIV, the swirling distribution around the gap-covered model in V1 is shown in Fig. 17. The vortex field around the gap-covered model is very similar to that of the static model (Fig. 10), both in scale and strength. The Kármán vortex street only exists in the wake. However, why is vibration not induced by wake vortex shedding on the gap-covered model? An important difference between the flow around the dynamic model and the gap-covered model is the location of vortex formation. In spite of the Kármán vortex street in the wake region of the gap-covered model, which provides periodic excitations, the vortex is not strong enough to magnify the oscillations due to the streamlined profile. The wake vortices have no direct action on the model, leading to a weak interaction between the vortex and the model. Therefore, it is the process (phase and timing) of vortex convection along the model surface that changes some crucial factors of the system, including the aerodynamic damping and the distribution of the VIV forces. Consequently, it leads to a strong coupling between the vortex and structure and the occurrence of VIV.

3.4. Vortex-shedding frequency

The vortex-shedding frequency (VSF) was measured with a hot wire anemometer in the wake zone of the static model and the dynamic model. The VSF power spectral density contours are normalized by maximum energy and shown in Fig. 18. For the static model, there is a main VSF at each velocity, and the frequency increases linearly with wind velocity, implying that the VSF follows a Strouhal law, and the Strouhal number is 0.236. Note that there is also a smaller frequency at $U/Df_v < 10$, and the corresponding St is 0.122. However, the energies of this St line remain at a low level and become invisible when U/Df_v is greater than 10. The results demonstrate that there can be only one Strouhal number playing a role in inducing vibrations.

For the dynamic model, the VIV ranges are also marked in the figures, from which we can see that ‘lock-in’ occurs in all four of the intervals, in which the vortex-shedding frequency remains the same as the natural frequencies. The energies are mainly focused on f_v and f_t in V1 and T1, with a small component of the doubling natural frequencies. While the energies of f_v and its double frequency are prominent in V2, and so are f_t and $2f_t$ in T2. The energy distributions imply that the vortex-shedding mode in V1 and T1 differs from that in V2 and T2. This is in line with the PIV measurements that indicated the vortex-shedding mode in V2 and T2 changed from the 2S mode to the 2(2S) mode.

The Strouhal frequency of the static model is exactly equal to the vertical natural frequency, f_v , at the beginning of V1, and 1.3 times of f_t when torsional vibration commences in T1. Therefore, the inducing factor for the VIVs occurrence in V1 and T1 is confirmed to be the resonance of the Strouhal frequency and the natural frequency. However, the Strouhal frequency in V2 is about

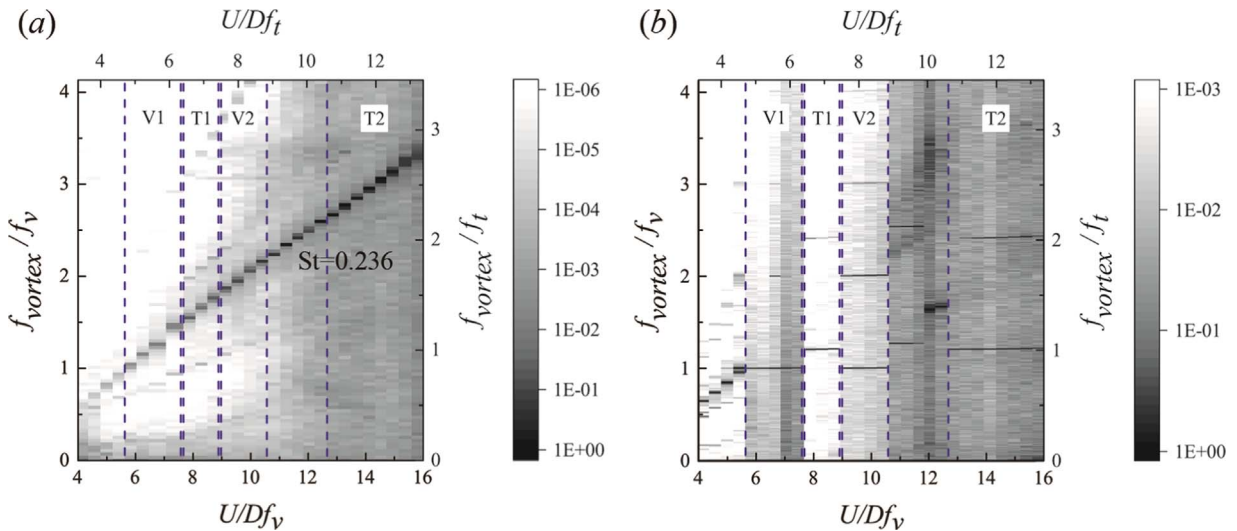


Fig. 18. Wake VSF power spectral density contours of (a) static model and (b) dynamic model.

twice of f_v , and rises from $2.2f_t$ to $2.8f_t$ in T2. Although the Strouhal frequency is far from the natural frequencies, the ‘lock-in’ happens anyway. It can be concluded that what happened in V2 and T2 is not just resonant but a 1:2 phase-locking between the oscillation and vortex-shedding, as proposed by Ermentrout (1981).

‘Lock-in’ is a type of synchronization in which the phase difference between vortex-shedding and oscillation is a constant, which can be indicated as $\varphi_{vortex} - \varphi = C$. Ermentrout (1981) mathematically studied the phase-locking phenomenon between two weakly coupled oscillators, and proved the existence of $n:m$ phase-locking. $n:m$ phase-locking implies $\varphi_A(nT_A) - \varphi_B(mT_B) = \psi \bmod 2\pi$, where φ_A and φ_B are the phases of two coupled oscillators A and B , T_A and T_B are the periods and $nT_A = mT_B$. The conditions and regions of phase-locking depend on the deviation of the natural frequency from $(n/m)\omega_B$ and the strength of coupling to the forcing function, where ω_B is the frequency of oscillator B .

The literature implies the possibility that the vibration of a model could be locked by vortex-shedding in which the frequency is an integral multiple of the natural frequency, when taking the VIV as the oscillator A and the vortex-induced force as the oscillator B . We believe that the special 1:2 mode ‘lock-in’ happened in V2 and T2.

4. Conclusions

Investigations of the process of vortex-induced vibration (VIV) are performed with a high mass ratio ($m^* = 904$) separated-box girder model. Vertical and torsional oscillations are induced in four wind velocity zones successively and repeatedly. The dynamical responses of the prototype and control groups certify that the central gap is essential for inducing vibrations, while wake vortex-shedding is not. Flow visualization measurements reveal the role of the central gap in changing the vortex field around the model. Comparison between the static and dynamic models shows a mutual-promotion relationship between the gap vortex and model motion. Vortices form around the gap and are strongly coupled with the structure during the leeward convection and VIV occurs. The flow interference at the gap can be disturbed by the oscillation of the model. The shear layer periodically rolls up and the Kármán vortex street appears in the gap, leading to the occurrence of wake-induced vibration (WIV). The WIV is induced by VIV and it magnifies the VIV in return. VIVs in the first two wind velocity intervals are caused by the Strouhal frequency reaching the natural frequency, while the VIVs in the last two intervals are due to a 1:2 phase-locking between the model motions and vortex-shedding. The vortex-shedding mode is observed to change from the 2S mode to the 2(2S) mode, in which there are two pairs of vortices shed in one vibrational cycle.

The central gap enables WIV to occur in V1, and it is also a crucial factor of VIVs in all four intervals, for it changes the location of vortex formation from the wake region to the leading corners of the downstream box girder. It can be concluded that it is the process of vortex convection along the surface of the downstream body that provides a higher level of coupling between the vortices and structure, and leads to the VIVs in all four velocity intervals.

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